

Step 8: Prediction Intervals for New Houses

Danish Housing Prices · Mathematical Statistics

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```
library(tidyverse)
dk_villa <- readRDS("../data/dk_villa.rds")
M_final <- readRDS("../data/M_final.rds")

cat("Model formula:", deparse(formula(M_final)), "\n")
cat("n =", nrow(dk_villa), "\n")
cat("s =", round(summary(M_final)$sigma, 5), "\n")
```

```
Model formula: log_price ~ log_sqm + house_age + nom_interest_rate. + dk_ann_infl_rate.
n = 402
s = 0.27463
```

1 Two Types of Interval

Both intervals share the same point estimate $\hat{\mu}(t_0) = \hat{\alpha} + \hat{\beta}^\top t_0$ but differ in what uncertainty they capture:

Confidence interval for the mean — uncertainty about where the true mean $\mu(t_0) = E[Y \mid t_0]$ lies:

$$\hat{\mu}(t_0) \pm t_{0.975}(n-p) \cdot s \cdot \sqrt{t_0^\top (X^\top X)^{-1} t_0}$$

Prediction interval for a single new observation — uncertainty about where one new Y_0 will fall, adding residual variance σ^2 :

$$\hat{\mu}(t_0) \pm t_{0.975}(n-p) \cdot s \cdot \sqrt{1 + t_0^\top (X^\top X)^{-1} t_0}$$

The prediction interval is always wider than the confidence interval by the extra 1 under the square root, reflecting irreducible observation noise.

2 New Houses to Predict

Three hypothetical Copenhagen Villas with characteristics representative of the 2023–2024 market:

```
# Adjust predictor names to match the final model's terms exactly
new_houses <- data.frame(
  log_sqm          = log(c(100, 130, 180)),
  house_age        = c(60, 50, 20),
  nom_interest_rate. = c(3.5, 3.5, 3.5),
  dk_ann_infl_rate. = c(3.3, 3.3, 3.3)
)

new_houses$sqm <- exp(new_houses$log_sqm)
print(new_houses)
```

```
      log_sqm house_age nom_interest_rate. dk_ann_infl_rate. sqm
1 4.605170      60      3.5              3.3 100
2 4.867534      50      3.5              3.3 130
3 5.192957      20      3.5              3.3 180
```

3 Computing the Intervals

```
ci_log <- predict(M_final, newdata = new_houses, interval = "confidence")
pi_log <- predict(M_final, newdata = new_houses, interval = "prediction")

sqm_labels <- c("100 sqm", "130 sqm", "180 sqm")

cat("=== Confidence intervals for E[Price | t0] ===\n\n")
for (i in 1:3) {
  cat(sprintf("  %s:  point est = %s DKK\n",
    sqm_labels[i],
    format(round(exp(ci_log[i,"fit"])), big.mark = ",")))
  cat(sprintf("          95%% CI = [%s - %s] DKK\n\n",
    format(round(exp(ci_log[i,"lwr"])), big.mark = ","),
    format(round(exp(ci_log[i,"upr"])), big.mark = ",")))
}
```

```

cat("=== Prediction intervals for a single new house ===\n\n")
for (i in 1:3) {
  cat(sprintf("  %s: 95%% PI = [%s - %s] DKK\n",
    sqm_labels[i],
    format(round(exp(pi_log[i,"lwr"]))), big.mark = ","),
    format(round(exp(pi_log[i,"upr"]))), big.mark = ",")))
}

```

=== Confidence intervals for $E[\text{Price} \mid t_0]$ ===

100 sqm: point est = 3,982,857 DKK
 95% CI = [3,743,711 - 4,237,279] DKK

130 sqm: point est = 4,422,537 DKK
 95% CI = [4,183,252 - 4,675,510] DKK

180 sqm: point est = 4,957,210 DKK
 95% CI = [4,589,659 - 5,354,195] DKK

=== Prediction intervals for a single new house ===

100 sqm: 95% PI = [2,313,009 - 6,858,231] DKK

130 sqm: 95% PI = [2,570,100 - 7,610,146] DKK

180 sqm: 95% PI = [2,873,307 - 8,552,489] DKK

4 Width Comparison

```

ci_widths <- exp(ci_log[, "upr"]) - exp(ci_log[, "lwr"])
pi_widths <- exp(pi_log[, "upr"]) - exp(pi_log[, "lwr"])

width_tbl <- data.frame(
  sqm          = c(100, 130, 180),
  point_est    = format(round(exp(ci_log[, "fit"])), big.mark = ","),
  CI_width_DKK = format(round(ci_widths), big.mark = ","),
  PI_width_DKK = format(round(pi_widths), big.mark = ","),
  PI_CI_ratio  = round(pi_widths / ci_widths, 1)
)
knitr::kable(width_tbl,
  col.names = c("sqm", "Point est (DKK)",

```

```

"CI width (DKK)","PI width (DKK)","PI/CI ratio"),
caption = "Interval widths on DKK scale")

```

Table 1: Interval widths on DKK scale

sqm	Point est (DKK)	CI width (DKK)	PI width (DKK)	PI/CI ratio
100	3,982,857	493,567	4,545,223	9.2
130	4,422,537	492,258	5,040,046	10.2
180	4,957,210	764,535	5,679,182	7.4

5 Plot: Predicted Price vs sqm

```

sqm_seq <- seq(60, 250, by = 2)
new_seq <- data.frame(
  log_sqm      = log(sqm_seq),
  house_age    = 55,
  nom_interest_rate. = 3.5,
  dk_ann_infl_rate. = 3.3
)

ci_seq <- predict(M_final, newdata = new_seq, interval = "confidence")
pi_seq <- predict(M_final, newdata = new_seq, interval = "prediction")

ylim_range <- range(exp(pi_seq[, c("lwr","upr")]))

plot(sqm_seq, exp(ci_seq[, "fit"]),
     type = "l", lwd = 2.5, col = "steelblue",
     ylim = ylim_range,
     main = "Predicted Price vs sqm - Copenhagen Villa\n(house_age=55, nom_rate=3.5%, infl",
     xlab = "sqm",
     ylab = "Price (DKK)")

# CI band
polygon(c(sqm_seq, rev(sqm_seq)),
        c(exp(ci_seq[, "lwr"]), rev(exp(ci_seq[, "upr"])))),
        col = rgb(0.27, 0.51, 0.71, 0.2), border = NA)

# PI band
polygon(c(sqm_seq, rev(sqm_seq)),

```

```

c(exp(pi_seq[, "lwr"]), rev(exp(pi_seq[, "upr"]))),
col = rgb(0.70, 0.13, 0.13, 0.1), border = NA)

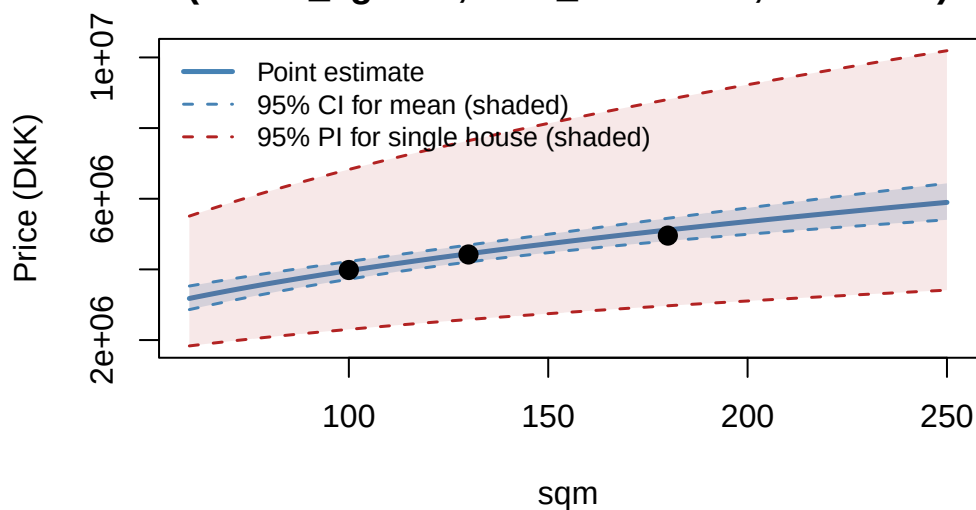
lines(sqm_seq, exp(ci_seq[, "lwr"]), lty = 2, col = "steelblue", lwd = 1.5)
lines(sqm_seq, exp(ci_seq[, "upr"]), lty = 2, col = "steelblue", lwd = 1.5)
lines(sqm_seq, exp(pi_seq[, "lwr"]), lty = 2, col = "firebrick", lwd = 1.5)
lines(sqm_seq, exp(pi_seq[, "upr"]), lty = 2, col = "firebrick", lwd = 1.5)

# Mark the three new houses
points(exp(new_houses$log_sqm), exp(ci_log[, "fit"]),
pch = 16, cex = 1.4, col = "black")

legend("topleft",
legend = c("Point estimate",
"95% CI for mean (shaded)",
"95% PI for single house (shaded)"),
col = c("steelblue", "steelblue", "firebrick"),
lty = c(1, 2, 2), lwd = c(2.5, 1.5, 1.5),
bty = "n", cex = 0.85)

```

Predicted Price vs sqm – Copenhagen Villa (house_age=55, nom_rate=3.5%, infl=3.3%)



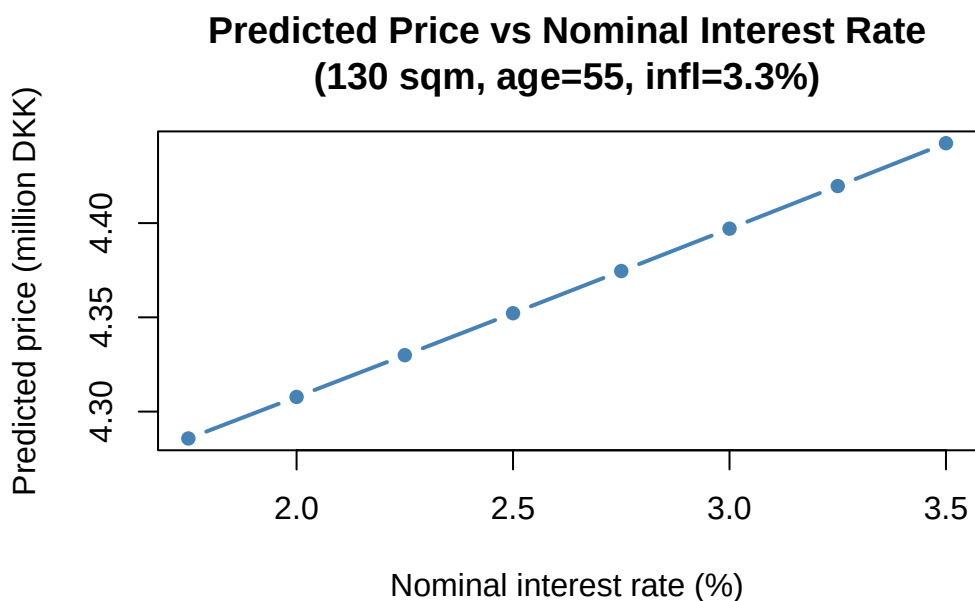
6 Sensitivity to Interest Rate

How does the predicted price change as `nom_interest_rate.` varies?

```
rates <- seq(1.75, 3.6, by = 0.25)
base_df <- data.frame(log_sqm      = log(130),
                      house_age   = 55,
                      nom_interest_rate. = rates,
                      dk_ann_infl_rate. = 3.3)

preds <- predict(M_final, newdata = base_df, interval = "confidence")

plot(rates, exp(preds[, "fit"]) / 1e6,
     type = "b", pch = 16, col = "steelblue", lwd = 2,
     main = "Predicted Price vs Nominal Interest Rate\n(130 sqm, age=55, infl=3.3%)",
     xlab = "Nominal interest rate (%)",
     ylab = "Predicted price (million DKK)")
lines(rates, exp(preds[, "lwr"]) / 1e6, lty = 2, col = "steelblue")
lines(rates, exp(preds[, "upr"]) / 1e6, lty = 2, col = "steelblue")
```



```
cat("Price at 1.75% rate:", format(round(exp(preds[1, "fit"])), big.mark=","), "DKK\n")
cat("Price at 3.60% rate:", format(round(exp(preds[nrow(preds), "fit"])), big.mark=","), "DKK\n")
cat("Difference:          ", format(round(exp(preds[1, "fit"]) -
                                         exp(preds[nrow(preds), "fit"])),
                                   big.mark=","), "DKK\n")
```

Price at 1.75% rate: 4,285,745 DKK
Price at 3.60% rate: 4,442,377 DKK
Difference: -156,632 DKK

7 Interpretation

CI vs PI — the key distinction:

A CI for the mean answers: “*What is the typical price of all 130 sqm villas with these characteristics?*” The CI narrows as $n \rightarrow \infty$ because we learn the mean precisely.

A PI for a new house answers: “*What will this specific house sell for?*” The PI never collapses to a point because individual transactions always carry irreducible noise σ^2 , no matter how large the dataset.

Back-transformation note: the intervals are symmetric on the log scale but asymmetric on the DKK scale (wider above the point estimate than below), correctly reflecting the log-normal structure of prices.